Phase 1: Classic ML with Scikit-learn & Web Deployment

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This notebook covers the full Phase 1 workflow: data loading, exploratory analysis, model training with Scikit-learn, evaluation, and export for browser-side prediction via ML.js.

The demo of the browser-side prediction is available on https://defineusernamexd.github.io/DataScienceExercise/phase1_deployment/.

Dataset: Steam Store Games (~27,000 games) **Task**: Regression – predict the retail price of a Steam game based on 10 features. **Target variable**: `price` (EUR)

In [8]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import json
```

1. Data Loading & Feature Engineering

The Steam dataset is loaded and 10 features are extracted from raw columns.

In [9]:
```
df = pd.read_csv("steam.csv")
def has_term(val, target):
    if not isinstance(val, str): return 0
    return 1 if any(t.strip().lower() == target.lower() for t in val.split(';')) else 0

df['release_year'] = pd.to_datetime(df['release_date'], errors='coerce').dt.year
df['release_year'] = df['release_year'].fillna(df['release_year'].median())
df['self_published'] = (df['developer'] == df['publisher']).astype(int)
df['is_mac'] = df['platforms'].apply(lambda x: 1 if isinstance(x, str) and 'mac' in x.lower() else 0)
df['is_multiplayer'] = df['categories'].apply(lambda x: has_term(x, 'Multi-player'))
df['is_indie'] = df['genres'].apply(lambda x: has_term(x, 'Indie'))
df['is_action'] = df['genres'].apply(lambda x: has_term(x, 'Action'))
df['is_early_access'] = df['genres'].apply(lambda x: has_term(x, 'Early Access'))

features = [
    'average_playtime', 'achievements', 'release_year', 'self_published', 'english',
    'is_mac', 'is_multiplayer', 'is_indie',
    'is_action', 'is_early_access'
]
target = 'price'

df_clean = df.dropna(subset=features + [target])
df_clean = df_clean[(df_clean['price'] >= 4.0) & (df_clean['price'] <= 80.0)]
df_clean = df_clean[~((df_clean['price'] <= 6.0) & (df_clean['achievements'] > 50))]

print(f"Dataset shape after cleaning: {df_clean.shape}")
print(f"Price range: €{df_clean['price'].min():.2f} – €{df_clean['price'].max():.2f}")
df_clean[features + [target]].describe()
```

```
Dataset shape after cleaning: (12727, 25)
Price range: €4.00 – €79.99
```

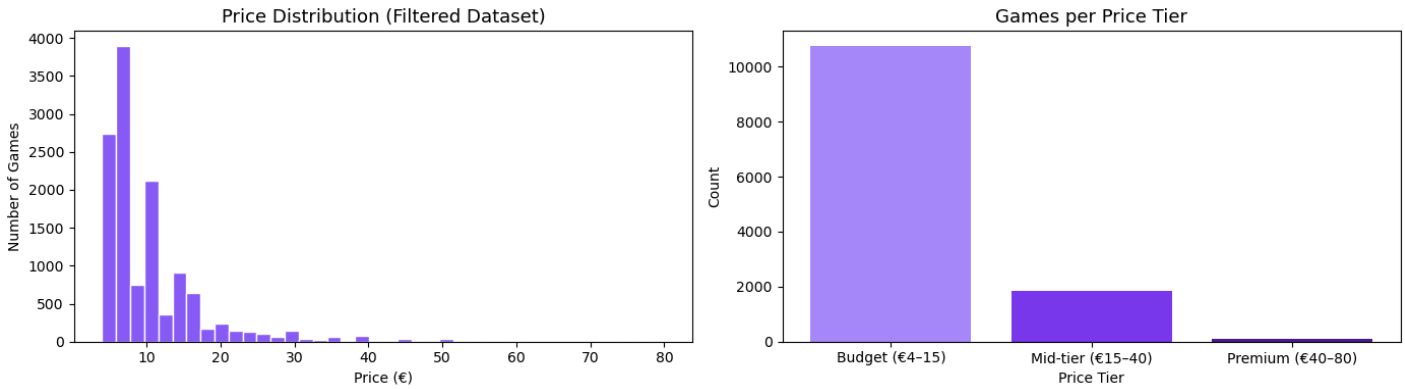
Out[9]:

	average_playtime	achievements	release_year	self_published	english	is_mac	is_multiplayer	is_indie	is
count	12727.000000	12727.000000	12727.00000	12727.000000	12727.000000	12727.000000	12727.000000	12727.000000	12727
mean	171.067180	35.546319	2016.14646	0.589141	0.985071	0.338572	0.187004	0.635028	0
std	1140.078512	288.627229	2.43057	0.492009	0.121273	0.473242	0.389930	0.481441	0
min	0.000000	0.000000	1997.00000	0.000000	0.000000	0.000000	0.000000	0.000000	0
25%	0.000000	0.000000	2015.00000	0.000000	1.000000	0.000000	0.000000	0.000000	0
50%	0.000000	12.000000	2017.00000	1.000000	1.000000	0.000000	0.000000	1.000000	0
75%	2.000000	28.000000	2018.00000	1.000000	1.000000	1.000000	0.000000	1.000000	1
max	55204.000000	9821.000000	2019.00000	1.000000	1.000000	1.000000	1.000000	1.000000	1

2. Exploratory Data Analysis (EDA)

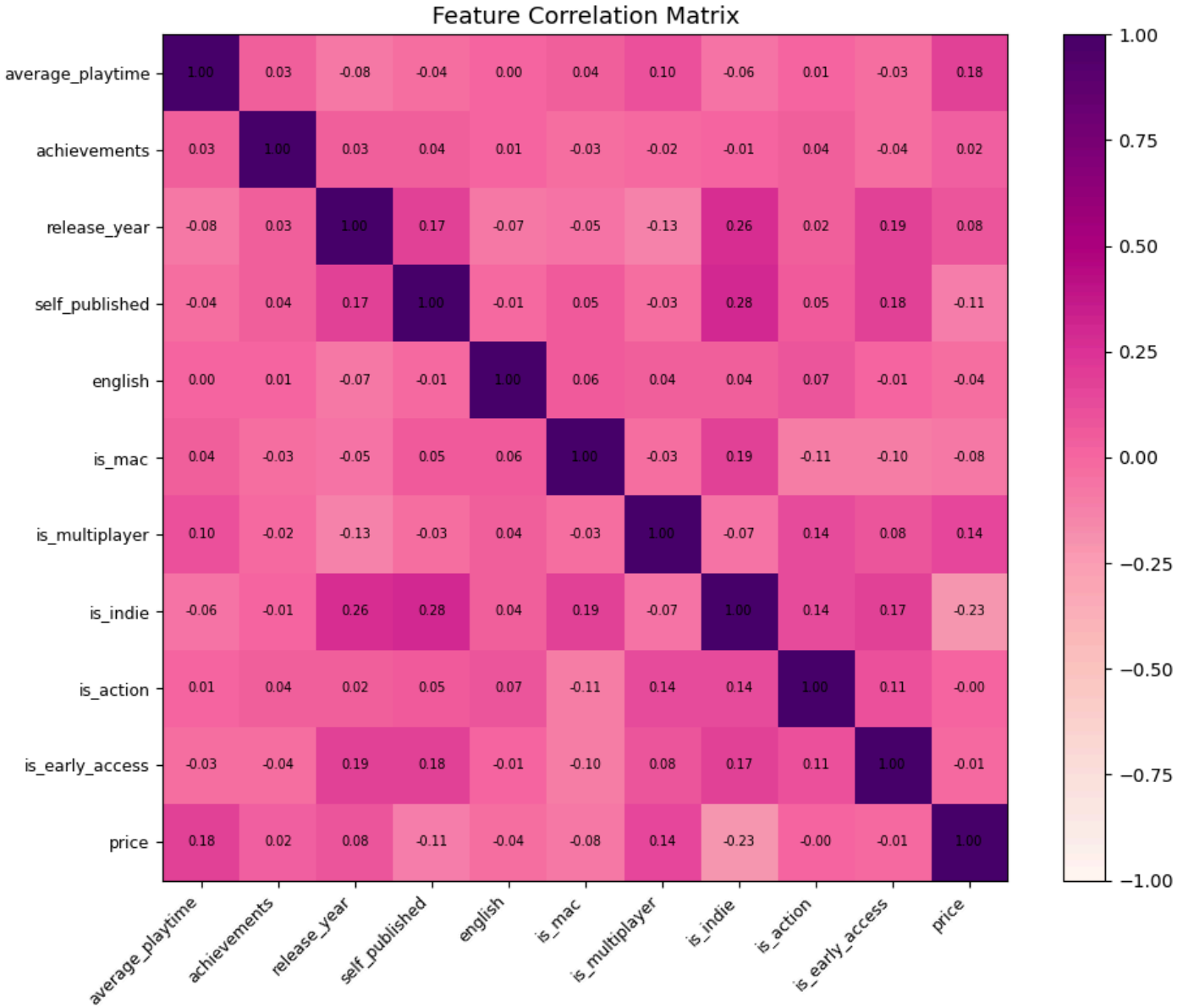
We examine the distribution of the target variable and the correlation structure of features before training.

In [10]:
```
# --- 2.1 Price Distribution ---
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

axes[0].hist(df_clean['price'], bins=40, color='#8b5cf6', edgecolor='white')
axes[0].set_title('Price Distribution (Filtered Dataset)', fontsize=13)
axes[0].set_xlabel('Price (€)')
axes[0].set_ylabel('Number of Games')

price_tier_counts = pd.cut(df_clean['price'], bins=[0, 15, 40, 80],
                           labels=['Budget (€4–15)', 'Mid-tier (€15–40)', 'Premium (€40–80)']).value_counts()
axes[1].bar(price_tier_counts.index, price_tier_counts.values, color=['#a78bfa', '#7c3aed', '#4c1d95'])
axes[1].set_title('Games per Price Tier', fontsize=13)
axes[1].set_xlabel('Price Tier')
axes[1].set_ylabel('Count')

plt.tight_layout()
plt.savefig('eda_price_distribution.png', dpi=120, bbox_inches='tight')
plt.show()
print("Most games cluster in the Budget tier (€4–15), confirming the strong indie presence on Steam.")
```




```
Most games cluster in the Budget tier (€4–15), confirming the strong indie presence on Steam.
```

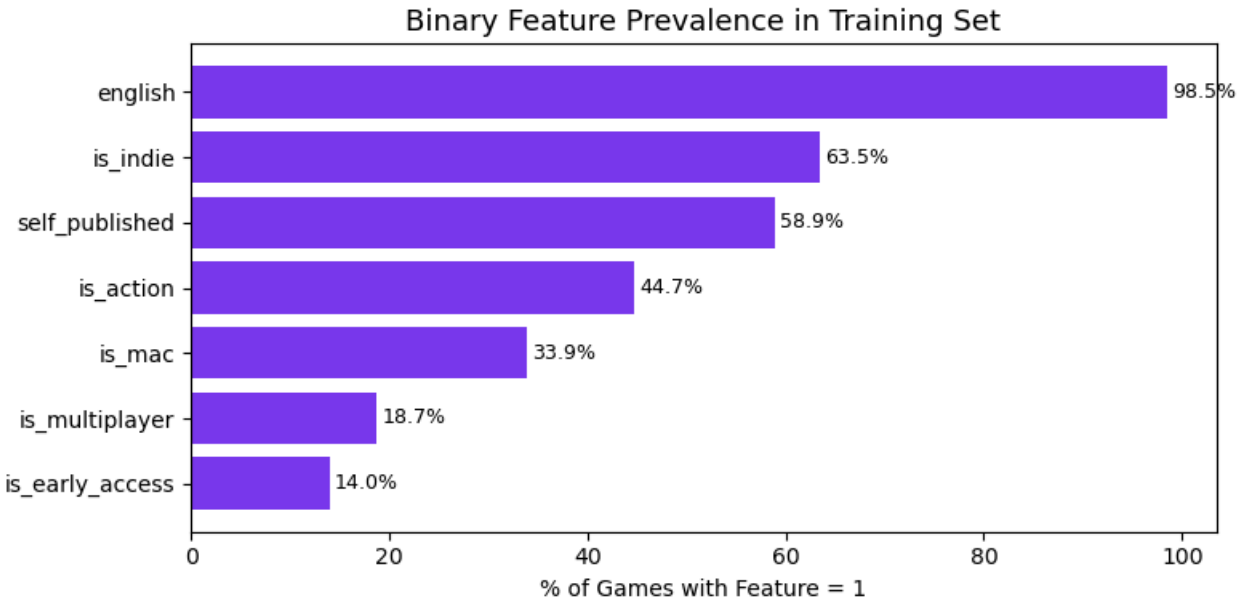
In [11]:
```
# --- 2.2 Correlation Heatmap ---
import matplotlib.colors as mcolors

corr = df_clean[features + [target]].corr()
fig, ax = plt.subplots(figsize=(10, 8))
cax = ax.imshow(corr, cmap='RdPu', vmin=-1, vmax=1)
fig.colorbar(cax)
ax.set_xticks(range(len(corr.columns)))
ax.set_yticks(range(len(corr.columns)))
ax.set_xticklabels(corr.columns, rotation=45, ha='right', fontsize=9)
ax.set_yticklabels(corr.columns, fontsize=9)
for i in range(len(corr)):
    for j in range(len(corr.columns)):
        ax.text(j, i, f'{corr.iloc[i, j]:.2f}', ha='center', va='center', fontsize=7)
ax.set_title('Feature Correlation Matrix', fontsize=13)
plt.tight_layout()
plt.savefig('eda_correlation.png', dpi=120, bbox_inches='tight')
plt.show()
print("Key: is_indie is negatively correlated with price (indie games are cheaper).")
print("average_playtime and achievements show mild positive correlation with price.")
```




```
Key: is_indie is negatively correlated with price (indie games are cheaper).
average_playtime and achievements show mild positive correlation with price.
```

In [12]:

```python
# --- 2.3 Binary Feature Prevalence ---
binary_features = ['self_published', 'english', 'is_mac', 'is_multiplayer',
                   'is_indie', 'is_action', 'is_early_access']
prevalence = df_clean[binary_features].mean().sort_values(ascending=True)

fig, ax = plt.subplots(figsize=(8, 4))
ax.barh(prevalence.index, prevalence.values * 100, color='#7c3aed')
ax.set_xlabel('% of Games with Feature = 1')
ax.set_title('Binary Feature Prevalence in Training Set', fontsize=13)
for i, v in enumerate(prevalence.values):
    ax.text(v * 100 + 0.5, i, f'{v*100:.1f}%', va='center', fontsize=9)
plt.tight_layout()
plt.savefig('eda_feature_prevalence.png', dpi=120, bbox_inches='tight')
plt.show()
```



3. Model Training & Validation

An 80/20 train-test split is used, stratified by price tier to ensure even representation of budget, mid-tier and premium games in both sets.

In [13]:
```
X = df_clean[features]
y = df_clean[target]
df_clean['price_tier'] = pd.cut(df_clean['price'], bins=[0, 15, 40, 100], labels=[1, 2, 3])

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=df_clean['price_tier']
model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
print(f"MAE:  {mean_absolute_error(y_test, y_pred):.2f} €")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, y_pred)):.2f} €")
print(f"R²:   {r2_score(y_test, y_pred):.3f}")
```

```
MAE:  4.52 €
RMSE: 6.67 €
R²:   0.102
```

4. Model Export

The trained model is exported as `model.json` containing the coefficients and intercept, which can be loaded directly in the browser via `ML.js`.

In [14]:
```
model_data = {
    "coef": model.coef_.tolist(),
    "intercept": float(model.intercept_),
    "feature_names": features
}
with open("model.json", "w") as f:
    json.dump(model_data, f)
print("model.json exported successfully.")
print(f"Features: {features}")
print(f"Intercept: {model.intercept_:.4f}")
print(f"Coefficients: {dict(zip(features, model.coef_.round(4)))}")
```

```
model.json exported successfully.
Features: ['average_playtime', 'achievements', 'release_year', 'self_published', 'english', 'is_mac', 'is_multiplayer', 'i
s_indie', 'is_action', 'is_early_access']
Intercept: -1063.7916
Coefficients: {'average_playtime': 0.001, 'achievements': 0.0002, 'release_year': 0.5349, 'self_published': -1.0511, 'engl
ish': -1.7395, 'is_mac': -0.3857, 'is_multiplayer': 2.554, 'is_indie': -3.5306, 'is_action': 0.1302, 'is_early_access': -
0.1008}
```